COMMERCIAL ITEM DESCRIPTION

CURING UNIT, DENTURE, TWO STAGE

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Federal Specification GG-C-860B which is cancelled.

This commercial item description covers a two-stage, automatically controlled curing unit, suitable for processing plastic denture material.

Salient Characteristics:

Fire and casualty hazards. The contractor shall submit proof to the contracting agency that the thermoswitch, on-off switch, strain relief, thermal cut-off, pilot light, wires, cord set, timer motor and timer switch he will supply under this specification conforms to the applicable requirements of the Underwriters Laboratory Inc. Standard UL 1262 for laboratory equipment. The label or listing of the Underwriters Laboratory Inc. may be accepted as evidence that they conform to this requirement.

Material.

Corrosion-resistant steel. Class 3, type 302, 303 or 304 austenitic stainless steel shall be in accordance with ASTM Designation F899 for composition only.

Design and dimensions. The curing unit shall be a two-stage, automatically controlled unit. The unit shall consist of an enclosure housing, tank cover, two grates, and necessary electrical components including a heating element, two timers, two thermostats, two pilot lights, and a power switch. The unit shall have the design and dimensions as shown in figure 1. The unit shall be capable of heating water in one or two stages to controlled temperatures in the range from 105°F to 212°F, for intervals of time from 15 minutes to 23 hours, depending on time settings and thermostat settings. The unit shall be capable of accommodating twelve denture flasks, in six flask presses, and up to fifteen flasks, by modifying the position of the grates. All electrical controls and necessary buss bars shall be installed on a non-metallic board mounted

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behind the front control panel with control shafts protruding through the control panel. The board shall be mounted in such a way as to provide easy access to all control components when the front panels are removed. The curing unit shall be designed to operate on a nominal 115 volt, 60 Hertz (Hz), single phase circuit.

Construction.

Enclosure housing. The enclosure housing shall have a sectional wrap-around, forming the two sides and rear, a one or three section front control panel, a baffle plate separating the controls section from the heating section, an electrical control mounting board, a bottom cover plate, and a drip board of the tank assembly which covers the control section. An opening shall be provided in the top of the enclosure to accept the water tank.

Front control panel. The control panel(s) shall be fabricated from cold-rolled steel having a thickness of 0.0598 inch (nominal). If a one piece control panel is supplied it shall be secured to the enclosure housing with a minimum of six corrosion-resistant steel machine screws. If a three section control panel is supplied, each panel shall be secured to the enclosure housing with a minimum of four corrosion-resistant steel machine screws. The control panel(s) shall be readily removable for servicing.

Control mounting board. The electrical controls mounting board shall be fabricated from a suitable material of high dielectric strength with appropriate mechanical properties. The board shall have a minimum thickness of 1/4 inch. The board shall be mounted to its supports by means of machine screws.

Baffle, wrap-around, bottom cover. The baffle, wrap-around and bottom cover shall be fabricated from cold rolled steel having a thickness of 0.0312 inch (nomimal).

Tank assembly. The tank and its dripboard shall be fabricated from corrosion-resistant steel. The tank portion shall be rectangular in shape having the following dimensions:

Width - 14 $\pm$ 1/4 inches

Length - 10 $\pm$ 1/4 inches

Depth - 12-1/2 $\pm$ 1/4 inches

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The tank shall have a minimum top opening of 13-5/8 inches by 9-5/8 inches. The thickness of the tank sides shall be 0.0312 inch (nominal). The thickness of the tank bottom shall be 0.050 inch (nominal). The dripboard section shall be welded to the peripheral top of the tank and shall provide drainage into the tank. The thickness of the dripboard shall be 0.0375 inch (nominal). All tank seams shall be welded as recommended by the American Welding Society. The tank shall be insulated on the sides and bottom with a minimum one (1) inch dead air space provided by the enclosure housing. The tank shall be equipped with a 3/8 inch (nominal) inside diameter brass drain pipe and a brass drain cock. The drain cock and visible portion of the drain pipe shall be finished with a baked black crinkle enamel or black texture vinyl coating. Sealing between the tank and corrosion-resistant drain and thermostat fitting shall be accomplished with copper gaskets.

Tank cover. The tank cover shall be fabricated from corrosion-resistant steel having a thickness of 0.0312 inch (nominal). The cover shall be provided with a molded plastic handle with a minimum 4 inch by 1 inch opening when assembled to the cover. Free edges of the cover shall be hemmed, and all corners shall be rounded. The cover shall be fitted, on the underside with a hollow flange frame to fit the opening of the tank.

Grate. The grates shall be cast in brass or bronze and nickel plated. The grates shall have four tapered legs, rounded corners and a minimum open grate area of 65 percent (%). The overall dimensions of the grate shall be 4-5/8 $\pm$ 1/8 inches by 13-5/8 $\pm$ 1/8 inches by 3-3/4 $\pm$ 1/8 inches high. The grates shall be designed to fit side by side within the tank to support the flask presses and to protect the heating element. The rear grate shall be designed so that when inverted with the legs extending upward, the three 3-flask presses instead of three 2-flask presses may be used, thereby increasing the total capacity of the curing unit to 15 flasks.

Heating element. The heating element shall be of the immersion type rated for 1650 watts at a nominal 115 volts, 60 Hz. The heating element shall be copper clad, nickel plated.

Timers. The timers shall consist of a nominal 115 volt, 60 Hz, single phase, synchronous type motor and the necessary mechanism for the timer to perform its intended function. The timers shall be provided with a screw-type terminal block for connecting the timers into the circuit. The timers shall be mounted to and behind the control panel, so that it is

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enclosed, and the dial is visible through a clear, plastic window on the control panel. The timer dials shall have a minimum diameter of 1-7/8 inches, and permanently and legibly marked from 15 minutes to 23 hours in intervals of 15 minutes. The timer contacts shall be protected by a relay.

Thermostat. The thermostats shall be of the immersion, strut-type construction, not bimetallic, enclosed in a corrosion-resistant steel sheath mounted to the tank by means of a threaded fitting. The controls shall be extended to the front control panel. The thermostats shall be readily removable for servicing. The thermostats' contacts shall be protected by a relay.

Temperature dials. The temperature dials shall be calibrated from 105°F to 212°F and permanently and legibly marked in increments of 20°F starting at 110°F mark. A metal pointer knob or other suitable means of indicating the temperature setting shall be provided.

Pilot lights. The pilot lights shall be of the neon type, with a nylon body. One assembly shall be supplied with a green lens (station 1) and the other assembly with a red lens (station 2).

Power switch. The power switch shall be a double pole, toggle switch, rated for 20 amperes at 115 volt, 60 Hz. The switch lever shall be clearly etched with the words "ON" and "OFF". The power switch shall serve only as a total disconnecting medium in the operation of the curing unit.

Power cord. The curing unit shall be equipped with 3-conductor power cord not less than 6 feet in length. The cord shall be a minimum of No. 14 AWG with a type SPT-3 thermoplastic insulation. The cord shall be equipped with a three prong plug suitable for use with receptacles specified in ANSI C73.11. The power cord shall enter the side of the unit through a strain relief fitting. The power cord shall conform to the requirements of Underwriters Laboratories, Inc. UL 1262 standard for laboratory equipment.

Performance.

Dielectric strength. The curing unit shall be capable of withstanding, without failure the dielectric strength test as specified in UL 1262.

Operational test. The curing unit shall operate as stated below. With the main power switch on and station 1 and

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station 2 set for a specific temperature and time interval both pilot lights shall be on. Add water to the tank to a depth of 10 ± 1/2 inches. Station 1 thermostat shall be set at 125°F and station 1 timer shall set at 3 hours. Station 2 thermostat shall be set at 212°F and station 2 timer shall be set at 1-1/2 hours. Turn on main switch. With the main power switch on and station 1 and station 2 set for the specific temperatures and time interval both pilot lights shall be on. When the preset time for station 1 has elapsed, the green pilot light shall go out and station 2 timer and thermostat shall take over for the remainder of the curing cycle. When the preset time for station 2 has elapsed the red pilot light shall extinguish and the curing unit shall turn off.

Heating rate. The heating element shall be capable of providing a uniform water temperature increase, in one or two stages of 2° ± 1/2°F. The following procedure shall be followed: Place four flask presses with two denture flasks each in the tank of the curing unit. Add sufficient water to cover the flasks. Set the timer and thermostat of station 1 to the "OFF" position. Set the timer and thermostat of station 2 to 90 minutes and 212°F respectively. Suspend a calibrated immersion type thermometer in the water. Cover the tank with the thermometer protruding thru. Record the temperature of the water. Turn the main power switch on. Monitor the time it takes the water temperature to read 200°F. Calculate the heating rate.

Heating element. The heating element shall be capable of withstanding continuous duty operation. The tank shall be filled with water. The station 1 thermostat shall be set for 150°F and station 1 timer for 12 hours. The station 2 thermostat shall be set at maximum and the station 2 timer shall be set for 23 hours. The main power switch shall be turned on and the curing unit permitted to operate for 24 hours. The water level shall be check occasionally and water added if necessary. The power consumption shall be monitored and recorded. The heater assembly shall show no signs of deterioration.

Timers. The timers shall maintain the set time within ± 2 percent. During performance of the heating element test, the timers shall be tested for accuracy with an accurate timepiece. Accurate recordings of the timepiece and the timers shall be made at the start of test and at the sixth, twelfth, eighteenth and twenty-third hour of the test period. The recorded data shall be used to determine the accuracy of the timers.

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Temperature control. The thermostat shall maintain the set temperature within ± 2 percent. The tank shall be filled water to a depth of 10 ± 1/2 inches. A calibrated immersion type thermometer shall be suspended in the water. The tank shall be covered. Each thermostat shall checked at intervals of 15°F starting at 105°F on station 1 thermostat and 110°F on station 2 thermostat. The water temperature shall be permitted to stabilize. The thermostat setting and thermometer reading shall be recorded at each interval.

Leakproof. The tank shall be free of leaks. At the completion of temperature control test the curing unit shall be checked for any water leakage from the tank. Any leakage shall be recorded.

Welding. Welding shall be durable, sound in every detail. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal of electrode where applicable to provide the best possible weld. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign substances. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut or excessive splatter. Sufficient soft metal shall be added to provide a suitable fillet of reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes or cracks.

Finish. All items fabricated of cold-rolled steel shall be suitably finished for protection against corrosion. The outside surface of the wrap-around shall be finished with baked black crinkle enamel or black texture vinyl coating. The tank, tank cover and dripboard shall have a satin finish. The bottom cover shall have a black enamel finish.

Service data. The contractor shall furnish, with each unit, two copies of a technical manual which contains complete instructions for installation, operation, and maintenance and a list of component parts.

Identification marking. Each curing unit shall be permanently and legibly marked with the following information: name or registered trademark of the manufacturer, manufacturer's model number, voltage, frequency and current. The serial number shall be die stamped into the surface of the tank at the upper right side, adjacent to the dripboard.

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Workmanship. The curing unit shall be free from defects such as burrs, feathers, sharp edges, pits, coarse grind marks, and cracks which detract from its appearance or impair its serviceability.

Unit. Each (EA). One curing unit, as specified, constitutes one unit. Complete and detailed instructions for use shall be supplied with each unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection, examination, and test characteristics. The unit of product for inspection, examination, test purposes shall be one curing unit. Unless otherwise specified, sampling for examination shall conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1, AQL 2.5 (percent defective) and sampling for tests (when applicable) shall be conducted at inspection level S-1. Acceptance number for tests shall be zero (0) for all samples, except Dielectric strength inspection level shall be 100 percent and any defect shall constitute rejection of the lot.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Chemical analysis. In the absence of contractor certification for the raw material, a chemical analysis shall be conducted. The chemical analysis shall be performed by reagent solutions, combustion methods, nonemission methods, or by spectrochemical analysis. Any ASTM method shall be acceptable.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-519-8000	CURING UNIT, DENTURE, TWO STAGE, 115 volt, 60 Hertz

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

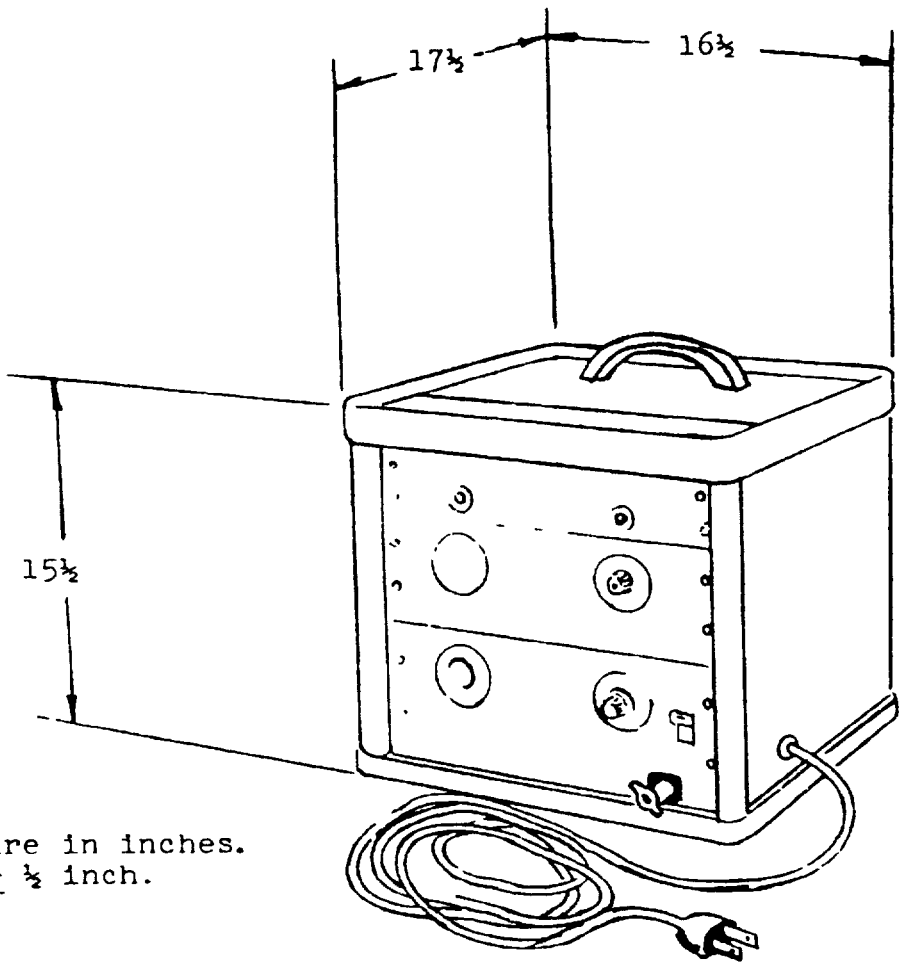
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2437
LOCATION: KAD1/curingu2

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Dimensions are in inches.
Tolerances ± ½ inch.

The illustration shown above is typical for style and design and is not intended to restrict or preclude the purchase of dental curing units which are otherwise in accordance with the requirements of this specification.

FIGURE 1. Curing unit, denture, two stage.

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54391	2. DOCUMENT DATE (YYMMDD) 4-30-91

3. DOCUMENT TITLE
Curing Unit, Denture, Two Stage

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center	(215)737-2870	444-2870
c. ADDRESS (Include Zip Code) ATTN: DPSC-MSTE 2800 S. 20th Street Philadelphia, PA 19101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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